Team Name : Argmax

Team Leader Name : Sathya T

Problem Statement : Rank the top-100 best-fit candidates from a 100,000-profile pool for the Senior AI Engineer (Founding Team) @ Redrob AI role - contextual + behavioral fit over keyword overlap, CPU-only and offline (<=5 min).

Solution Overview

What is your proposed solution?

- A hybrid, explainable, CPU-only ranking engine: honeypot filter -> JD gated rubric -> semantic + lexical relevance -> cross-encoder rerank -> behavioral multiplier -> fact-grounded reasoning.

What makes it different from keyword matching?

- Substance over keywords: scores what candidates BUILT (career history); the skills list is excluded, so keyword-stuffers don't rank.
- Reads profiles: an impossibility/honeypot filter removes logically-impossible profiles (0 in our top-100 vs the >10% that disqualifies).
- Availability-aware: a bounded behavioral multiplier down-weights unreachable 'perfect-on-paper' candidates.
- Compliant by design: heavy models run offline; the scored step is stdlib, CPU-only, offline, ~190s.

JD Understanding & Candidate Evaluation

Key requirements extracted from the JD

- Must-have: production embeddings retrieval, vector/hybrid search, strong Python, ranking evaluation (NDCG/MRR/MAP/A-B).
- Ideal profile: 5-9 yrs, product company (not pure services/research), shipped an end-to-end ranking/search/recsys system, Pune/Noida.
- Negatives: services-only, CV/speech-without-NLP, title-chasers, recent-LLM-only.

Evaluating fit beyond keyword matching

- Career-evidence of retrieval/ranking work + an engineering-role gate ('Marketing Manager' + AI skills ~ 0).
- Skill DEPTH (proficiency x endorsements/duration x assessments), not count; embeddings catch plain-language Tier-5s; 23 Redrob signals weigh availability.

Ranking Methodology

Retrieve -> score -> rerank

- Recall funnel: a fast structured score over all 100k keeps a top pool.
- Semantic: bge-small-en-v1.5 cosine to a distilled JD query, blended 50/50 with lexical concept matching.
- Rerank: bge-reranker-v2-m3 cross-encoder takes half-authority over the top tier (drives NDCG@10).
- Weighted components: relevance .30, title/career .28, skill-depth .20, experience .12, education .10.
- Then multiply by soft penalties (services/CV-speech/title-chaser/location) and a bounded behavioral availability factor (0.5 to 1.15).
- Final: score = fit x penalties x behavioral; tie-break by candidate_id. Models run offline; rank.py reads cached JSON.

Explainability & Data Validation

How decisions are explained

- Each candidate shows a component breakdown + relevance parts (lexical / semantic / cross-encoder) and a 1-2 sentence reasoning.

No hallucinations

- Reasoning is templated strictly from real profile facts (no LLM), so it cannot cite anything not in the profile; tone matches rank.

Suspicious profiles

- Honeypot filter (e.g. 'expert' skill with 0 months used). 65 flagged, 0 false positives, 0 reach the top-100.

Explainability in the demo UI

End-to-End Workflow

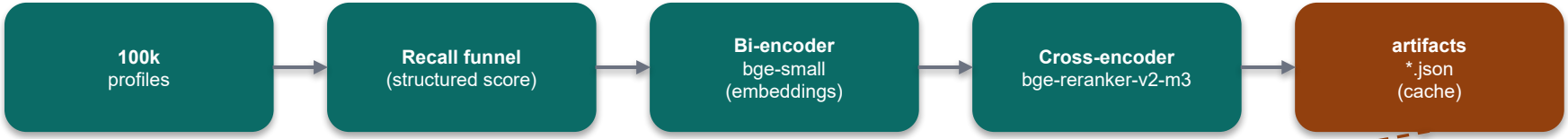
Complete workflow: JD input -> ranked output

- OFFLINE (GPU ok, one-time): embed the candidate pool + cross-encoder rerank -> cache artifacts/*.json.
- SCORED (rank.py, CPU/offline, <=5 min): stream 100k -> honeypot filter -> component scoring x behavioral multiplier -> top-100 heap -> fact-grounded reasoning -> submission.csv.
- Validate: data/validate_submission.py confirms format before upload.

Full workflow diagram: docs/05_approach_and_roadmap.md

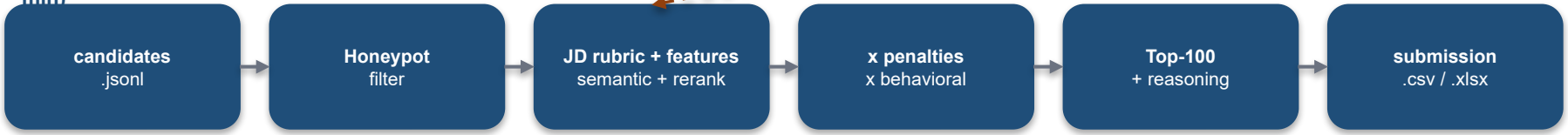
System Architecture

1 · OFFLINE PRECOMPUTE (GPU, one-time)



cached scores -> loaded by rank.py

2 · SCORED RANKING - rank.py (CPU · offline · <=5 min)



Demo sandbox: app/ (FastAPI) + frontend/ (React + Tailwind) -> single Docker image (HF Space), reading the same src/. Compliance: GPU / transformers live only in precompute; rank.py is stdlib · CPU · offline.

Results & Performance

0
honeypots in top-100

~190s
CPU · offline run

83/100
in the 5-9 yr band

97/100
show ranking work

```
                    $ just check        # rank.py -> submission.csv, then validate
Scored 100,000 candidates (65 honeypots filtered) in 186.8s; top score 1.03, cutoff 0.77
Wrote 100 ranked candidates to submission.csv     Submission is valid.
$ uv run python scripts/quality_proxy.py submission.csv
top-100:  in-band 83/100   ranking-work 97/100   juniors 1/100   available 92/100
```

Top-10: senior product-company retrieval/ranking/NLP engineers (Meta, Apple, Netflix, Zomato, Paytm, Razorpay). Ground truth hidden - validated by ablation + calibration (docs/09).

Technologies Used

Technologies & why

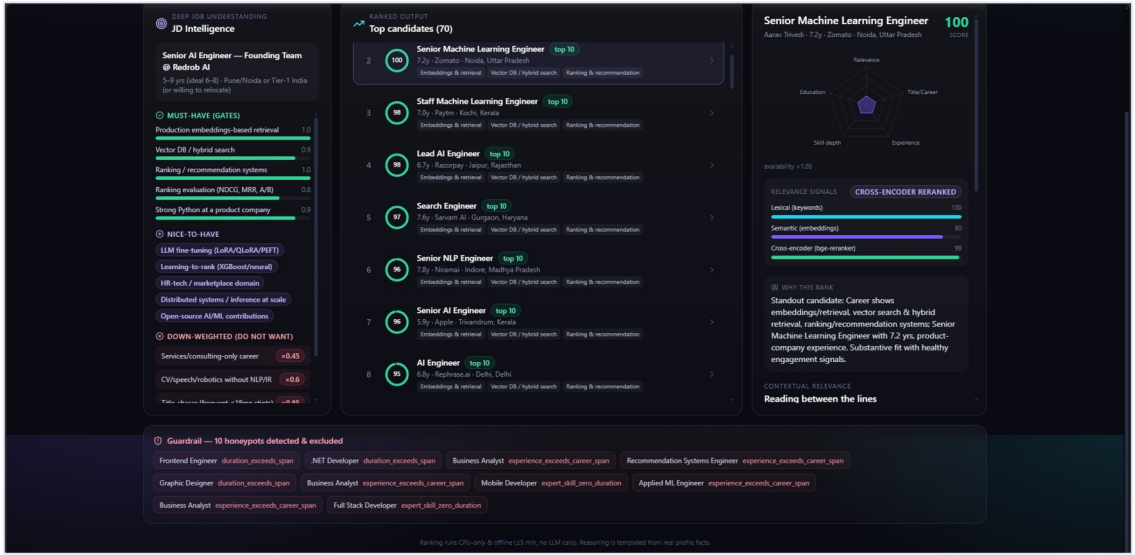
- Ranker: Python 3.13, stdlib-only - deliberate, guarantees CPU/offline reproducibility at Stage 3.
- Offline precompute: PyTorch + sentence-transformers; bge-small-en-v1.5 (embeddings) + bge-reranker-v2-m3 (cross-encoder) - open-source, no hosted API (Cohere excluded: network).
- Demo: FastAPI; Vite + React + TypeScript + Tailwind; Recharts.
- Tooling: uv (reproducible envs, no pip); just (cross-platform tasks); Docker (HF Space).

Every choice keeps the scored path inside the compute budget while heavy models run offline.

Submission Assets

GitHub (public) · ranked output XLSX · approach deck (PDF) · docs/00-09 + PROJECT_LOG

LIVE on Hugging Face Space · tsathya98-redrob-candidate-ranker.hf.space



redrob | H2S HACK2SKILL

INDIA.RUNS

Build what next India runs on

THANK YOU